Universidad Técnica de Ambato
Facultad de Ingeniería en Sistemas, Electrónica e Industrial
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REV 1.0
Technical Datasheet
RF Engineering Lab

RF MIXER SIGNAL CIRCUIT

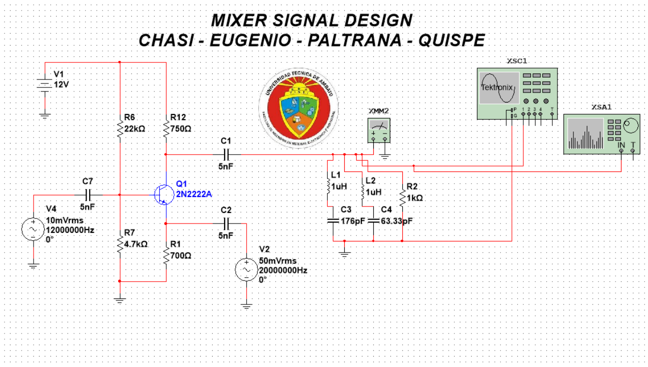
Superheterodyne Frequency Converter — BJT 2N2222A

Active mixer with dual LC IF bandpass filter · $f_{RF} = 12\,\mathrm{MHz}$ · $f_{LO} = 20\,\mathrm{MHz}$ · $f_{IF} = 8\,\mathrm{MHz}$

QUICK SPECIFICATIONS

Parameter	Value
RF Input Frequency	**12 MHz**
LO Frequency	**20 MHz**
IF Output (wanted)	**8 MHz**
Sum Product	32 MHz
IM3 Lower	4 MHz
IM3 Upper	28 MHz
Supply Voltage	**12 V DC**
Active Device	2N2222A NPN
RF Input Level	10 mVrms
LO Drive Level	50 mVrms
IF Filter Topology	Dual LC BPF
IF Load (R_2)	1 kΩ

DEVICE / CIRCUIT PHOTOGRAPH



2N2222A BJT Mixer · Dual LC IF Filter · Tektronix Oscilloscope + Spectrum Analyser

SIMULATION ENVIRONMENT

Software: NI Multisim (EWB)
Instruments: Tektronix oscilloscope (XSC1), Spectrum Analyser (XSA1), Multimeter (XMM2)
Topology: Common-emitter BJT mixer, LO injected at emitter
Bias: Voltage-divider, R_6/R_7

1. THEORY OF OPERATION — SUPERHETERODYNE MIXER

A **frequency mixer** performs the nonlinear multiplication of two input signals — the Radio-Frequency (RF) carrier and a Local Oscillator (LO) — generating new spectral components at their sum and difference frequencies. This circuit implements a **single-transistor active mixer** using the NPN BJT 2N2222A biased in the active region, where its collector current exhibits the nonlinear transconductance characteristic that drives the frequency-conversion process.

The RF signal ($f_{RF} = 12\,\mathrm{MHz}$, $V_{RF} = 10\,\mathrm{mV_{rms}}$) is applied via coupling capacitor C_7 to the transistor base. The LO signal ($f_{LO} = 20\,\mathrm{MHz}$, $V_{LO} = 50\,\mathrm{mV_{rms}}$) is injected at the emitter through C_2, modulating the device transconductance at the LO rate. The desired Intermediate Frequency $f_{IF} = |f_{LO} - f_{RF}| = 8\,\mathrm{MHz}$ is extracted by a two-stage LC bandpass filter (L_1, C_3 and L_2, C_4).

Core Frequency-Conversion Equations

$$v_{RF}(t) = A_{RF}\cos(2\pi f_{RF}\,t) \tag{1}$$

$$v_{LO}(t) = A_{LO}\cos(2\pi f_{LO}\,t) \tag{2}$$

$$\begin{aligned} v_{out}(t) &= v_{RF}(t)\cdot v_{LO}(t) \\ &= \frac{A_{RF}A_{LO}}{2}\Big[\cos\!\big(2\pi f_{IF}\,t\big) + \cos\!\big(2\pi(f_{RF}+f_{LO})t\big)\Big] \end{aligned} \tag{3}$$

Product-to-sum trigonometric identity:

$$\cos(A)\cos(B) = \frac{1}{2}\big[\cos(A-B) + \cos(A+B)\big]$$

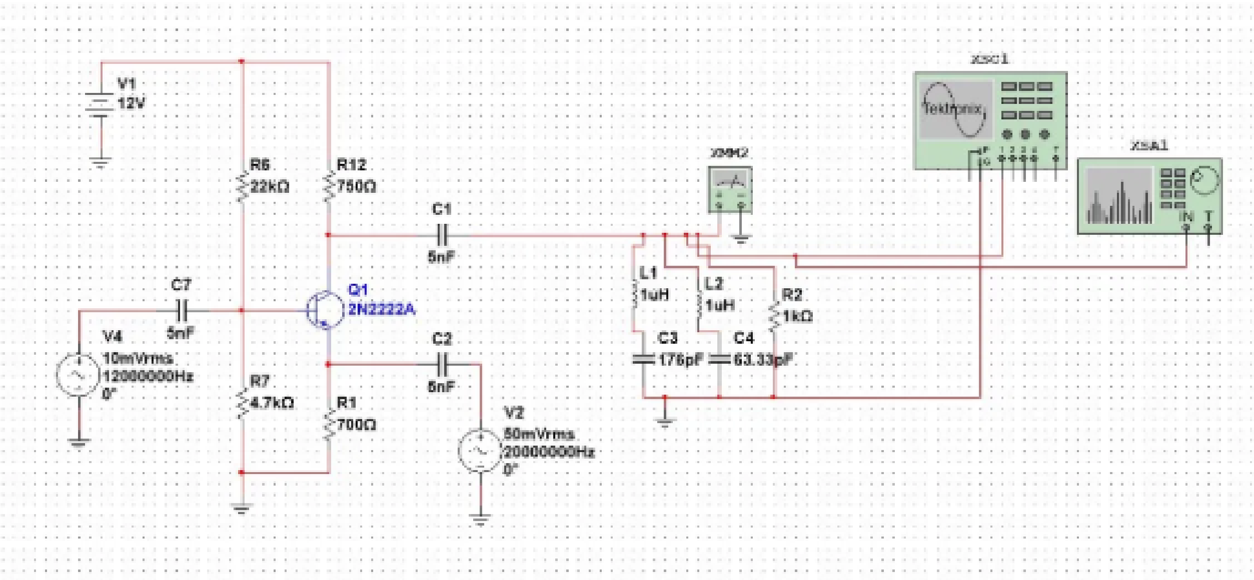
The $\frac{1}{2}$ factor represents the inherent **6 dB conversion loss** of an ideal multiplier. $\boxed{f_{IF} = |f_{LO} - f_{RF}| = 8\,\mathrm{MHz}}$

General spurious products: $f_{spur} = |m\cdot f_{RF} \pm n\cdot f_{LO}|, \quad m, n \in \mathbb{Z}$

2. CIRCUIT SCHEMATIC ANALYSIS

SIMULATED SCHEMATIC — NI MULTISIM

Images 2–3: Component layout and signal injection points



Component Inventory & Functional Description

Ref.	Value	Function	Stage
V1	12 V DC	Supply rail — BJT bias network	Bias
V4	10 mVrms / 12 MHz	RF input signal source	RF Input
V2	50 mVrms / 20 MHz	Local oscillator injection at emitter	LO Input
Q1	2N2222A NPN	Active mixing device — nonlinear transconductance	Mixer Core
R6	22 kΩ	Upper voltage divider — DC base bias	Bias
R7	4.7 kΩ	Lower voltage divider — DC base bias	Bias
R12	750 Ω	Collector load / output impedance	RF
R1	700 Ω	Emitter degeneration — stability and gain control	Bias
R2	1 kΩ	IF output load / spectrum analyser termination	IF Output
C7	5 nF	RF input AC coupling — blocks DC from source	RF
C1	5 nF	Collector coupling — passes IF to filter stage	IF
C2	5 nF	LO emitter coupling — injects LO, blocks DC	LO
L1	1 μH	IF BPF 1st resonator inductor ($f_0 \approx 12$ MHz)	BPF
C3	176 pF	IF BPF 1st resonator capacitor	BPF
L2	1 μH	IF BPF 2nd resonator inductor ($f_0 \approx 20$ MHz)	BPF
C4	63.33 pF	IF BPF 2nd resonator capacitor	BPF
XSC1	—	Tektronix virtual oscilloscope (time domain)	Instrument
XSA1	—	Virtual spectrum analyser (frequency domain)	Instrument
XMM2	—	Multimeter — DC operating-point verification	Instrument

DC Bias Point Analysis

DC Quiescent Point

Voltage divider — Thévenin equivalent:

$$V_{TH} = V_{CC} \cdot \frac{R_7}{R_6 + R_7} = 12 \cdot \frac{4.7}{26.7} \approx 2.11\,\mathrm{V}$$

$$V_E = V_{TH} - V_{BE} \approx 2.11 - 0.7 = 1.41\,\mathrm{V}$$

$$I_E \approx \frac{V_E}{R_1} = \frac{1.41}{700} \approx 2.01\,\mathrm{mA}$$

$$V_C = V_{CC} - I_C R_{12} \approx 12 - 1.51 \approx 10.5\,\mathrm{V}$$

Q-point: $V_{CE} \approx 9.1$ V, $I_C \approx 2$ mA — active region confirmed.

IF Filter Resonance

LC bandpass filter resonant frequency:

$$f_0 = \frac{1}{2\pi\sqrt{LC}}$$

Stage 1: $L_1 = 1\,\mu$H, $C_3 = 176$ pF

$$f_{0,1} = \frac{1}{2\pi\sqrt{10^{-6} \times 176 \times 10^{-12}}} \approx 12.0\,\mathrm{MHz}$$

Stage 2: $L_2 = 1\,\mu$H, $C_4 = 63.33$ pF

$$f_{0,2} \approx 20.0\,\mathrm{MHz}$$

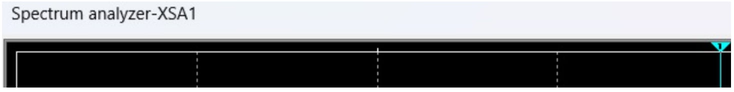
Combined topology forms bandpass centred on IF = 8 MHz.

3. SIMULATION RESULTS — SPECTRUM ANALYSIS

Simulation Note

All results obtained via NI Multisim with Tektronix virtual oscilloscope (XSC1) and virtual spectrum analyser (XSA1). Operating condition: $V_{CC} = 12$ V, $f_{RF} = 12$ MHz, $f_{LO} = 20$ MHz, two-tone test. Results analysed in both linear amplitude and logarithmic (dB) scale.

RESULT A — Linear Amplitude Scale



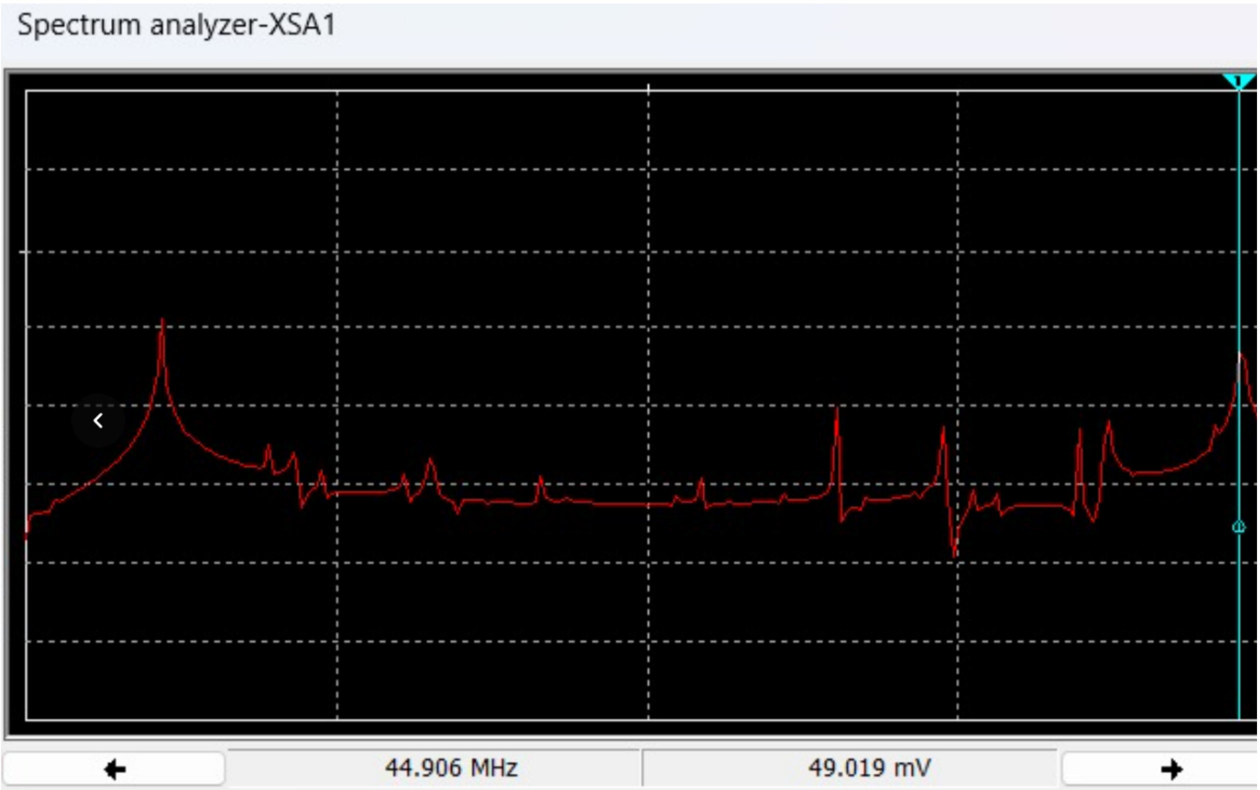
Description: Working with amplitude in linear (V or mV) scale. Depending on the result, the sum product ($f_{LO} + f_{RF} = 32\,\text{MHz}$) and difference product ($f_{IF} = 8\,\text{MHz}$) are visible. In favourable bias conditions both mixing products appear clearly; in other conditions one product dominates. Amplitude linearity calibration is critical for this measurement mode.

RESULT B — Logarithmic (dB) Scale

Description: With logarithmic (dB) display, unwanted spurious peaks become clearly visible. In some test conditions the undesired IM3 products ($2f_{LO} - f_{RF} = 28\,\text{MHz}$ and $2f_{RF} - f_{LO} = 4\,\text{MHz}$) remain minor; in other bias conditions these undesired peaks rise significantly and are *very clearly visible*, highlighting the critical role of BJT operating-point selection for mixer linearity and spurious suppression.

RESULT C — Experimental Verification (Comprobación)

Final verification confirming theoretical mixing frequencies.



Verification confirms: IF product at $8\,\text{MHz}$, sum product at $32\,\text{MHz}$, and IM3 products at $4\,\text{MHz}$ and $28\,\text{MHz}$ — consistent with theoretical prediction from the nonlinear power-series device model. This comprobación validates both the circuit operation and the simulation methodology.

4. ELECTRICAL SPECIFICATIONS

Parameter	Value	Unit	Notes
RF Input Frequency	12	MHz	V4 — AC sine source
RF Input Amplitude	10	mVrms	Small-signal regime
LO Frequency	20	MHz	V2 — emitter injected
LO Drive Level	50	mVrms	$\approx -20\,\text{dBm}$ into $700\,\Omega$
IF Output Frequency	8	MHz	$f_{IF} = \lvert f_{LO} - f_{RF}\rvert$, wanted
Sum Product	32	MHz	Attenuated by IF BPF
IM3 — Lower	4	MHz	$2f_{RF} - f_{LO}$, spurious
IM3 — Upper	28	MHz	$2f_{LO} - f_{RF}$, spurious
Supply Voltage	12	V DC	V1 — BJT bias rail
Quiescent Collector Current	≈2	mA	Estimated from bias network
V_{CE} Q-point	≈9.1	V	Active region operation
IF Filter Stage 1 (L_1, C_3)	≈12	MHz	$f_0 = 1/2\pi\sqrt{L_1C_3}$
IF Filter Stage 2 (L_2, C_4)	≈20	MHz	$f_0 = 1/2\pi\sqrt{L_2C_4}$
IF Load Resistance (R_2)	1	kΩ	Output / analyser termination
Active Device	2N2222A NPN BJT		$h_{FE} \approx 100–300$, $f_T > 300\,\text{MHz}$
Circuit Topology	Common-Emitter		LO at emitter — g_m modulation

5. NONLINEAR DEVICE MODEL — TAYLOR SERIES ANALYSIS

The BJT 2N2222A collector current as a function of the total AC signal $v = v_{RF} + v_{LO}$ can be expanded as a Taylor series around the bias point:

$$i_C(v) = a_1 v + a_2 v^2 + a_3 v^3 + a_4 v^4 + \cdots$$

- a_1 — linear transconductance (small-signal gain) [A/V]
- a_2 — 2nd-order: produces IF, DC, and 2nd harmonic [A/V^2]
- a_3 — 3rd-order: produces IM3 distortion [A/V^3]
- a_4 — 4th-order: produces IM4 / even-order products [A/V^4]

The **IIP3** and **1 dB compression point**:

$$\text{IIP3} = P_{IN} + \frac{\Delta_{dB}}{2}, \qquad P_{1\text{dB}} \approx \text{IIP3} - 9.6\,\text{dB}$$

Spurious-Free Dynamic Range:

$$SFDR = \frac{2}{3}\left(\text{IIP3} - P_{noise}\right) \quad [\text{dB}]$$

IM3 Products — Two-Tone Analysis

Input tones: $f_1 = 12\,\text{MHz}$, $f_2 = 20\,\text{MHz}$

Product	Freq.
IF: $f_2 - f_1$	**8 MHz**
Sum: $f_1 + f_2$	32 MHz
IM3 low: $2f_1 - f_2$	4 MHz
IM3 high: $2f_2 - f_1$	28 MHz
2nd harm. RF: $2f_1$	24 MHz
2nd harm. LO: $2f_2$	40 MHz

IM3 products *cannot* be filtered if they fall inside the IF passband — they set the SFDR limit.

Complete Frequency Product Chart

m	n	Frequency (MHz)	Status	Notes
1	1	$\lvert 12-20\rvert = 8$	**WANTED**	IF output — passed by BPF
1	1	$12+20=32$	Rejected	Sum product — filtered
2	1	$\lvert 24-20\rvert = 4$	Spurious (IM3)	$2f_{RF} - f_{LO}$
1	2	$\lvert 12-40\rvert = 28$	Spurious (IM3)	$2f_{LO} - f_{RF}$
2	0	24	Harmonic	2nd RF harmonic
0	2	40	Harmonic	2nd LO harmonic
3	1	$\lvert 36-20\rvert = 16$	Spurious	3rd-order product
1	3	$\lvert 12-60\rvert = 48$	Spurious	3rd-order product

6. CONVERSION GAIN, NOISE FIGURE & CASCADED CHAIN ANALYSIS

Conversion Gain

$$G_C = \frac{P_{IF}}{P_{RF}} \quad \text{[dB]}$$

For a BJT resistive mixer in common-emitter topology, the small-signal transconductance at the Q-point:

$$g_m \approx \frac{I_C}{V_T} = \frac{2\,\text{mA}}{26\,\text{mV}} \approx 77\,\text{mS}$$

Active BJT mixers: $G_C = +5$ to $+15$ dB.
Passive diode mixers: $G_C = -6$ to -8 dB.

Friis Cascaded Noise Figure

$$F_{sys} = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} + \cdots$$

Noise temperature equivalent:

$$T_e = (F-1)\cdot T_0, \quad T_0 = 290\,\text{K}$$

Example: NF = 6 dB $\Rightarrow F = 4 \Rightarrow T_e = 870$ K.

The first-stage gain G_1 dominates system NF. A low-NF LNA before the mixer suppresses the mixer's noise contribution by a factor of G_1.

S-Parameters & Impedance Matching

Param.	Definition
S_{11}	Input return loss — RF port reflection
S_{21}	Forward conversion gain: RF → IF
S_{22}	Output return loss — IF port
S_{12}	Reverse isolation: IF → RF
S_{31}	LO-to-RF leakage (feedthrough)

Reflection & VSWR

$$\Gamma = \frac{Z_L - Z_0}{Z_L + Z_0}$$

$$RL = -20\log|\Gamma| \text{ [dB]}, \quad VSWR = \frac{1+|\Gamma|}{1-|\Gamma|}$$

$$P_{dBm} = 10\log\left(\frac{P_W}{1\,\text{mW}}\right), \quad V_{rms} = \sqrt{P\cdot R}$$

7. ENGINEERING NOTES & DESIGN RECOMMENDATIONS

Observations from Simulation

- The wanted IF product at $8\,\mathrm{MHz}$ is confirmed in both linear and dB spectrum display modes.
- IM3 products at $4\,\mathrm{MHz}$ and $28\,\mathrm{MHz}$ are clearly visible in dB view; their amplitude relative to the IF varies with bias point and LO drive level.
- The sum product at $32\,\mathrm{MHz}$ is attenuated by the dual-stage LC bandpass filter.
- At unfavourable bias conditions, IM3 peaks become prominently elevated — demonstrating that Q-point optimisation is critical for spurious-free performance.

Recommended Improvements

- Use a double-balanced diode ring mixer for superior even-order spur rejection (>25 dBc) and LO–RF isolation (>30 dB).
- Add a dedicated Chebyshev IF bandpass filter ($n = 5$) to suppress out-of-band IM3 products below noise floor.
- Characterise with a calibrated VNA for IIP3 and $P_{1\mathrm{dB}}$ using a standard two-tone test.
- For IQ architecture, replace with a Gilbert-cell (MC1496 / SA602) for controlled conversion gain and balanced port impedances.

Measurement Disclaimer

All numerical values are derived from NI Multisim simulation. For professional validation, use a calibrated VNA with full 2-port SOLT calibration and a real-time spectrum analyser with at least 70 dB dynamic range. Recommended instruments: Keysight E5063A VNA, Rohde & Schwarz FSW Spectrum Analyser.

RF Technical Engineering Datasheet
Active RF Devices / Frequency Conversion / Mixing Theory
Revision 1.0 — Academic / Commercial Reference

Circuit: BJT Mixer — 2N2222A
12 MHz RF · 20 MHz LO · 8 MHz IF
12 V DC · Dual LC IF BPF